

Conclusions

This thesis presented the experimental realization and numerical analysis of the subsystems required to trap, transport, and cool a sample of ^{87}Rb atoms inside a Hollow-Core Photonic Crystal Fiber.

The preliminary cooling stages were characterized by optimizing the optical molasses, reaching a minimum temperature of $T = (8.42 \pm 0.17) \mu\text{K}$. The transfer of the atomic cloud into the Optical Dipole Trap was then investigated. By translating the MOT position via the compensation coils to improve the overlap with the dipole beam, a capture efficiency of $(0.161 \pm 0.005)\%$ was measured, resulting in about 10^5 trapped atoms.

The generation of the optical conveyor belt requires a second, counter-propagating 1064 nm optical field. The injection line for a polarization-maintaining single-mode fiber was built and aligned. The optical fields were overlapped on the upper breadboard to form the standing wave. To evaluate the reliability of the lattice potential, the stability of the transmitted field was measured, yielding a minimum fractional power deviation of 8×10^{-4} at 0.1 s and an angular polarization deviation of 7×10^{-5} rad at 3 s. While long-term stability is limited by low-frequency drifts, the measured parameters are compatible with the transport requirements.

To address the heating introduced by the transport process, the Electromagnetically Induced Transparency (EIT) cooling scheme was investigated numerically. A 24-level quantum master equation model was implemented to simulate the dynamics of the ^{87}Rb D_2 line. The results show that optical pumping into uncoupled states halts the cooling cycle, making the addition of an active repump field a requirement to maintain the population in the active Λ system and approach the motional ground state.

The integration of these subsystems provides the optical infrastructure to perform the dynamic transport of atoms into the HCPCF. The numerical analysis of EIT cooling also establishes the operating parameters for the future implementation of sub-Doppler cooling inside the hollow core.